RevKit v0.2.0

Feature and Use Guide for Reviewer Feedback

A browser-based workspace for planning, conducting, reading, analysing, and exporting systematic reviews.

SYSTEMATIC REVIEWS - META-ANALYSIS - EVIDENCE SYNTHESIS

What to test	Open an example review, add a study from PDF/DOI/link, attach a full-text PDF, try QUADAS-3 or RoB, export PRISMA and review files.
Best reviewers	Systematic reviewers, evidence synthesis methodologists, DTA reviewers, statisticians, librarians, and clinical content experts.
Current status	Functional prototype. It is suitable for expert workflow feedback, not yet a validated production replacement for established review platforms.

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1. What RevKit Is

RevKit is an open-source review workspace designed to keep common systematic-review tasks in one place: protocol preparation, reference and study setup, full-text reading, data entry, meta-analysis, risk-of-bias assessment, PRISMA flow management, and export.

The current interface has been remade as a compact premium dashboard with glass-style panels, a single prominent create-review call to action, responsive mobile layout, and a cleaner first screen so reviewers can start or resume work without scrolling through repeated wizard cards.

The aim is not to automate judgement away from reviewers. RevKit tries to remove avoidable clicking, copying, and tab switching while keeping the reviewer in control of each field and decision.

2. Current Review Types and Example Reviews

Review type	**Bundled example**	Use
Intervention	Aspirin after myocardial infarction	Dichotomous outcome data, intervention pooling, RoB 2 / ROBINS-I workflows.
Diagnostic test accuracy	Rapid antigen tests for influenza	2 x 2 data, DTA plots, QUADAS-3 workspace, diagnostic protocol structure.
Methodology	Machine-assisted study screening	Methods-review framing and effect estimates for review-method comparisons.
Overview of reviews	Exercise for chronic low back pain	Generic inverse-variance synthesis and AMSTAR/ROBIS guidance context.
Flexible / qualitative	Remote primary care experiences	Custom review structure for qualitative or mixed evidence.

3. Easier Study Setup

- Smart Add Study accepts a PDF file, DOI, or pasted webpage link and fills the study fields it can identify, such as title, authors, year, DOI, design notes, index test, reference standard, and notes.
- The reviewer checks and edits the fields before adding the study; metadata parsing is a helper, not an automatic final decision.
- PDF import is limited to files up to 15 MB for smart parsing. Study-level PDF attachment supports larger files up to 250 MB for local reading.
- DOI and link parsing depend on available public metadata and may need manual correction.

4. Full-Text Reading Inside RevKit

RevKit now includes an in-app full-text reader so reviewers do not need to keep switching between browser tabs, file viewers, and extraction forms.

- Attach, replace, open, and download a PDF from the Studies page.
- Read the PDF inside RevKit with previous/next study navigation.
- Keep reading notes beside the PDF; notes are saved back to the study record.
- Use browser full-screen where supported for focused reading.
- PDFs are stored in browser IndexedDB on the same device. They are not synced to other devices and may be lost if browser storage is cleared.

5. Protocol, PICO/PIRD, and PROSPERO Preparation

The protocol builder uses plain-language questions to help a reviewer prepare a registration-ready draft. For DTA reviews it uses PIRD-style thinking: population, index test, reference standard, and diagnosis/target condition.

- Sections cover purpose, eligibility, methods, and registration administration.
- Information helpers explain what each field means, why it matters, and show a practical example.
- A registration readiness bar tracks essential fields.
- The DTA sepsis template adds seven planned strata: adult, pregnant, paediatric, and neonatal populations with blood culture or consensus reference standards as relevant.
- Review-scoped team members can populate the PROSPERO team field, so one review's team does not automatically become every other review's team.
- The downloaded PROSPERO draft is a Word-compatible working document. It must still be checked against the live PROSPERO form and current word limits before submission.

AMSTAR 2 guidance is included where it fits. It is useful for overviews or reviews of systematic reviews, while primary diagnostic-accuracy studies should use QUADAS-3.

6. Risk of Bias and Applicability

- DTA reviews now prioritise QUADAS-3 v1.2 for estimate-level risk of bias and applicability.
- QUADAS-3 workspace includes synthesis questions, ideal test accuracy trial framing, study flow, estimates, domain assessments, and overall judgements.
- QUADAS-2 is preserved as a legacy option only for older protocols that explicitly require it.
- Intervention and methodology reviews can use RoB 2 and ROBINS-I with signalling questions, computed overall judgement, traffic-light plot, and summary bar plot.

7. Data Entry, Meta-analysis, and Plots

- Data types include dichotomous 2 x 2, continuous mean/SD, O-E and V, generic inverse variance, and diagnostic 2 x 2 TP/FP/FN/TN data.
- Effect measures include RR, OR, RD, Peto OR, MD, SMD, and DOR depending on data type.
- Pooling methods include Mantel-Haenszel, Peto, inverse variance, and DerSimonian-Laird random effects.
- Heterogeneity outputs include Q, I2, tau2, and H.
- Plots include intervention forest plots, DTA sensitivity/specificity forest plots, SROC plots, and funnel plots.
- The DTA calculator reports sensitivity, specificity, PPV, NPV, LR+, LR-, prevalence, DOR, and confidence intervals from entered 2 x 2 values.

8. PRISMA, Export, and Local Workspace

- PRISMA 2020 flow uses an 11-box template with automatic counts from review state or manual override.
- Export options include Word-compatible review output, CSV data export, and plot/image export where available.
- Reviews and PDFs are designed around a local browser workspace at this stage. There is no real-time multi-user editing or account-based sync yet.
- The app uses a cleaner RevKit logo and removes external model/vendor branding from the user interface.

9. Suggested Reviewer Test Script

- Open the dashboard and confirm the main call to action, saved reviews panel, and examples are understandable on desktop and phone.
- Choose the example closest to your own work and inspect whether the tabs match how you would conduct that review.
- Add one study from a PDF, DOI, or URL and check whether the filled fields are helpful and safe to edit.
- Attach a PDF, read it in RevKit, write notes, and move to another study.
- For a DTA review, open QUADAS-3 and assess whether the phases, domain wording, and estimate-level structure feel right.
- Try the protocol builder and download a PROSPERO draft. Check whether the language would help a junior reviewer prepare a registration.
- Export PRISMA, CSV, plots, or review output and note anything missing for real submission or publication workflows.

10. Known Limits and Safeguards

- RevKit is not a medical decision tool and bundled examples are illustrative only.
- Smart parsing is a convenience feature. Reviewers must verify all extracted metadata and manually complete fields that the parser cannot infer.
- PDFs stored locally in the browser are not backed up automatically unless future sync or account storage is configured.
- The PROSPERO draft is preparation material, not an official submitted registration.
- Real-time collaboration, user authentication, cloud document storage, audit trails, and production-grade permissions are future work.

11. Feedback Requested

- Are the workflow names and field labels clear to reviewers who do not know RevKit?
- Does Smart Add Study reduce busy work without giving a false sense of automation?
- Does the in-app PDF reader make extraction and notes easier enough to keep?
- Is QUADAS-3 presented in a way that matches current DTA appraisal practice?
- Would the PICO/PIRD and PROSPERO helpers make registration easier for a real protocol team?
- What methodological checks or export details would block real-world use?